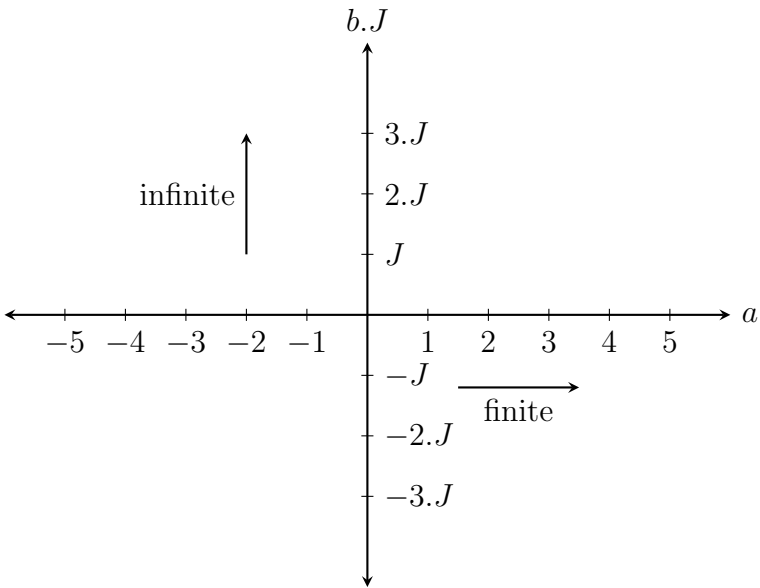


# THE J-MATH

Let  $\frac{1}{0} = J$

Let there be a 2D space, such that:



It is expressed in the form of

$$“a + bJ” \rightarrow \text{J-numbers}$$

$$\{a, b \in \mathbb{R}\}$$

$$\begin{aligned} a &= \text{finite part} \\ b.J &= \text{infinite part} \end{aligned}$$

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$0 \cdot (a + bJ) = b$

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## Addition

Addition is performed **component-wise**. Let there be two J-numbers,  $x$  and  $y$ , such that:

$$\begin{aligned}x &= a + bJ \\ y &= c + dJ\end{aligned}$$

Then, the sum is:

$$\begin{aligned}x + y &= (a + bJ) + (c + dJ) \\ x + y &= (a + c) + (b + d)J\end{aligned}$$

## Exponential/Power Rules

Since  $J = \frac{1}{0}$ , we establish the following rules:

1. For  $n > 0$ :

$$J^n = \left(\frac{1}{0}\right)^n = J = \frac{1}{0}$$

2. For  $n = 0$ :

$$J^0 = \left(\frac{1}{0}\right)^0 = 1$$

3. For  $n < 0$ :

$$J^n = \left(\frac{1}{0}\right)^n = 0$$

## Multiplication [Piecewise]

Let there be two J-numbers,  $x$  and  $y$ , such that:

$$\begin{aligned}x &= a + bJ \\ y &= c + dJ\end{aligned}$$

Then, the product  $x \cdot y$  is defined as:

$$x \cdot y = \begin{cases} b & \text{if } y = 0 \quad [\text{i.e., } c = 0 \text{ and } d = 0] \\ d & \text{if } x = 0 \quad [\text{i.e., } a = 0 \text{ and } b = 0] \\ ac + (ad + bc + bd)J & \text{otherwise} \end{cases}$$

## Standard Real Numbers as J-Numbers

All standard real numbers can be written as J-numbers, such that their J-coefficient ( $b$ ) is zero.

$$\begin{aligned}0 &= 0 + 0J \\ 7 &= 7 + 0J \\ -5 &= -5 + 0J\end{aligned}$$

## Important Note

One thing to be noted is that:

$$0 \cdot J \neq 0J$$

OR

$$(0 + 0J) \cdot (0 + 1J) \neq (0 + 0J)$$

## NOTE

1. **Commutativity** is intact.
2. **Distributive property** is abandoned conditionally. Its conditions are as follows:

$$X \cdot (Y + Z) \neq X \cdot Y + X \cdot Z$$

This applies if and only if all these following conditions are met:

- The multiplier ( $X$ ) has a non-zero J-coefficient ( $b$ ).
- The inside terms ( $Y + Z$ ) sum to exactly absolute zero.

Otherwise, the distributive property holds and is intact.

3. **Associative property** is abandoned conditionally. Its conditions are as follows:

$$(X \cdot Y) \cdot Z \neq X \cdot (Y \cdot Z)$$

If and only if all these following conditions are met:

- An Absolute zero is involved.
- A J-number is involved.
- That J-number has a non-zero coefficient ( $b$ ).

Otherwise, the Associative property holds and is intact.

## The Universal Golden Formula

### 1st way

Let there be a function of  $x$ , such that it is in the form of  $a + bJ$  for some variable  $x$ :

$$F(x) = A + Bx$$

where  $\{A, B, x \in \text{J-numbers}\}$

Let:

$$A = a_1 + a_2J$$

$$B = b_1 + b_2J$$

Let  $x$  be a variable in the form of  $a + bJ$ .

**Evaluate**  $F(J)$

$\Rightarrow$  substitute  $x \rightarrow 0 + 1J$

$$F(0 + 1J) = [a_1 + a_2J] + [(b_1 + b_2J) \cdot (0 + 1J)]$$

Using rules of multiplication established earlier:

$$\begin{aligned} F(0 + 1J) &= [a_1 + a_2J] + [b_1(0) + (b_1(1) + b_2(0) + b_2(1))J] \\ F(0 + 1J) &= [a_1 + a_2J] + [0 + (b_1 + 0 + b_2)J] \end{aligned}$$

Using rules of addition established earlier:

$$F(0 + 1J) = (a_1 + a_2J) + (b_1 + b_2)J$$

$$\boxed{F(0 + 1J) = a_1 + (a_2 + b_1 + b_2)J} \quad (1)$$

$F(x) = A + Bx$  is a linear J-function.

OR

Since in the function

$$F(x) = A + Bx$$

where  $A$  and  $B$  are some constants, we find the values of  $A$  and  $B$ .

$\Rightarrow$  put  $x = 0$

$$F(0) = A + B(0) = A + b_2$$

Hence:

$$\boxed{A = F(0) - b_2} \quad \text{--- (1)} \quad (2)$$

$\Rightarrow$  put  $x = 1$

$$F(1) = A + B(1)$$

Then:

$$B = F(1) - A$$

From Eq. (1) put  $A = F(0) - b_2$ :

$$B = F(1) - [F(0) - b_2]$$

Hence:

$$\boxed{B = F(1) - F(0) + b_2} \quad \text{--- (2)} \quad (3)$$

From Eq. (1) & (2) we have values of  $A$  and  $B$  now.

$\Rightarrow$  for Evaluation at  $J$

$$F(J) = A + BJ$$

From Eq. (1) & (2):

$$\boxed{F(J) = (F(0) - b_2) + (F(1) - F(0) + b_2)J} \quad (4)$$

For ordinary functions where  $b_2 = 0$

[**Important note** – All ordinary functions are subset of J-Functions], we get:

$$\boxed{F(J) = F(0) + [F(1) - F(0)]J} \quad (5)$$

## 2nd way

Let there be a function of  $x$ , such that:

$$F(x) = \sum_{n=0}^{\infty} C_n x^n$$

$$\text{At } F(0) = C_0(1)$$

$$\text{At } F(1) = C_0 + C_1 + C_2 + C_3 + \dots$$

$$\text{Now, } F(1) - F(0) = C_1 + C_2 + C_3 + \dots (2)$$

## Evaluation at $x = J$

Put  $x = J$  in  $F(x) = \sum_{n=0}^{\infty} C_n x^n$ :

$$F(J) = C_0 + C_1 J^1 + C_2 J^2 + C_3 J^3 + C_4 J^4 + \dots$$

Since  $J^n = J$  for  $n > 0$ :

$$F(J) = C_0 + C_1 J + C_2 J + C_3 J + C_4 J + \dots$$

$$F(J) = C_0 + (C_1 + C_2 + C_3 + C_4 + \dots)J$$

From Eq (1) and Eq (2):

$$\boxed{F(J) = F(0) + [F(1) - F(0)]J} \quad (6)$$

## 3rd way

Let there be a function of  $x$  such that:

$$F(x) = A + Bx$$

$\Rightarrow$  A Linear J-function.

Using the Taylor Series:

$$F(x) = \sum_{n=0}^{\infty} \frac{F^{(n)}(a)}{n!} (x - a)^n$$

For our work, we center the series at  $a = 0$ :

$$F(x) = \sum_{n=0}^{\infty} \frac{F^{(n)}(0)}{n!} x^n$$

Using the Taylor series for our Linear J-function:

$$F(x) = A + Bx$$

Because this function is linear, the 1st derivative is constant ( $F'(x) = B$ ) and all higher-order derivatives are zero.

Now, this becomes:

$$\boxed{F(x) = F(0) + F'(0)x} \quad \text{--- (1)} \quad (7)$$

$$F'(0) = \frac{F(x_2) - F(x_1)}{x_2 - x_1}$$

In our standard anchors  $x_2 = 1$  and  $x_1 = 0$ , we get:

$$\begin{aligned} F'(0) &= \frac{F(1) - F(0)}{1 - 0} \\ F'(0) &= F(1) - F(0) \quad \text{--- (2)} \end{aligned} \tag{8}$$

Put Eq. (2) in Eq. (1):

$$F(x) = F(0) + [F(1) - F(0)]x$$

Since at  $x = J$ , it becomes  $x = 0 + 1J$ :

$$F(0 + 1J) = F(0) + [F(1) - F(0)](0 + 1J)$$

OR

$$\boxed{F(J) = F(0) + [F(1) - F(0)]J} \tag{9}$$

## Sequence and Series Analysis

### Sequence Analysis

Let there be a sequence  $a_n$ , such that it is treated as a function of  $n$ :

$$f(n) = a_n$$

Then, using the UGF, the infinite-th term is:

$$\begin{aligned} f(J) &= f(0) + [f(1) - f(0)]J \\ \boxed{f(J) &= a_0 + [a_1 - a_0]J} \end{aligned} \tag{10}$$

### Series Analysis (Sum of Sequences)

Let there be (sum of sequences) i.e. a series, such that, its sum up to the term  $n$  is and it is treated as a function:

$$S(n) = \sum_{k=0}^n a_k$$

Then, using the UGF:

$$\begin{aligned} S_J &= S(0) + [S(1) - S(0)]J \\ S_J &= a_0 + [(a_0 + a_1) - a_0]J \\ \boxed{S_J &= a_0 + a_1J} \end{aligned} \tag{11}$$

## Absolute Zero

Absolute Zero =  $0 + 0J$  [i.e., the origin]

Since  $J = \frac{1}{0}$ ,  $J$  represents infinity.

Since  $J$  is the inverse of zero, then zero is the inverse of  $J$ :

$$\text{If } J = \frac{1}{0} = 0^{-1}$$

$$\text{Then } 0 = \frac{1}{J} = J^{-1}$$

## Origins of Infinity

Since  $5J$  is different from  $2J$  (i.e.,  $5(\text{infinity})$  is different from  $2(\text{infinity})$ ):

$$\text{Then } 5J \neq 2J$$

An infinity originated from 5 is different from an infinity originated from 2. J's (or infinities) remember their origin.

## Types of Infinities and Zeros

There are different types of infinities (J's). The same is valid for zero ( $J^{-1}$ ):

$$5J^{-1} \neq 2J^{-1}$$

i.e.,  $5(\text{zero})$  is different from  $2(\text{zero})$ .

Then J-inverses (or zeros) remember their origin:

$$5J^{-1} \neq 2J^{-1} \quad [\text{Information wise}]$$

A zero originated from 5 is different from a zero originated from 2.

$\therefore$  There are different types of Zeros (J-inverses)

## Absolute Infinity

$$\text{Absolute Infinity} = 0 + 1J$$

**NOTE:** While these equations evaluate to the same Absolute Zero, or the same Absolute Infinity, the lineage, or history of how they got there, is different. *Information is preserved.*

## The Generalized Universal Golden Formula

The generalized UGF is defined as:

$$\boxed{F(a + bJ) = F(a) + [F(a + b) - F(a)]J} \quad (12)$$

## Binomial Expansion in J-Space

Let there be a binomial expansion such that:

$$(a + bJ)^n$$

Then it is equal to:

$$(a + bJ)^n = a^n + \left[ \sum_{k=1}^n \binom{n}{k} a^{n-k} b^k \right] J \quad (13)$$

**Example:**  $\rightarrow f(x) = x^2$

Substitute  $x = a + bJ$ :

$$(a + bJ)^2 = a^2 + b^2 J^2 + 2abJ$$

Since  $J^n = J$  for  $n > 0$ , we get:

$$\boxed{(a + bJ)^2 = a^2 + (2ab + b^2)J} \quad (14)$$

## Taylor Expansion for $J^{-1}$

Since:

$$\frac{1}{0} = J = 0^{-1}$$

Then:

$$\frac{1}{J} = 0 = J^{-1}$$

Using the Taylor expansion series, we similarly get:

$$F(x + J^{-1}) = F(x) + \left[ F'(x) + \frac{F''(x)}{2!} + \frac{F'''(x)}{3!} + \dots \right] J^{-1} \quad (15)$$

## The Nature of Equality

Equality in J-Math is different. Equality is contextual rather than absolute. Context matters in J-Math.

### Types of Equality in J-Math $\Rightarrow$

1. **Destinational Equality**  $\Rightarrow$  When two things are equal in value [i.e., in destination] but have different J-coordinates.  
 $Eg \rightarrow \pi = \pi$  [whether evaluated using Leibniz's formula or Ramanujan's formula]
2. **Full Entity Equality**  $\Rightarrow$  When two things are equal in value [i.e., in destination] as well as their J-coordinates are the same.  
 $Eg \rightarrow \pi(\text{Ramanujan}) \neq \pi(\text{Leibniz})$  in context of full entity equality.

End of Framework Definition